PAPER_1203_UQFF_Canonical_v1.5_Simultaneous_Solver_Convergence

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PAPER_1203 -- UQFF Canonical v1.5 Simultaneous Solver Convergence (F_U=0)

Canonical v1.5 Master Equation (QCalcGeom.py:11, MAIN_1:2878) ``` F_U_total = (Ug1 + Ug2 + Ug3 + Ug4) - FUBi + FUBii + Um = 0 F_U = Ug_sum - Ubi + Um + dissipation Equilibrium at every shell/scale via FUBi(r) + FUBii(r) = 0 ```

Simultaneous Equation System (QCalcGeom.py:14) ``` Eq1: FUBi(r, t_n) + FUBii(r, t_n) = 0 # buoyancy crossing / compaction Eq2: \varepsilon'(r, t_n) + G*M/(c^2*r^2) = 0 # metric-geodesic -> r_hz, t_n_hz (HabitableZoneCalculator.solve_habitable_zone) ```

FUBi/FUBii Definitions (QCalcGeom.py:445, _uqff_primitives.py:766) ``` FUBi = -beta(t) * G * M * rho / r2 * (1+F_TRZ) * |cos(pi*t_n)| FUBii = +beta(t) * (r/r0) * k_spring * (1 + E_n) * |cos(pi*t_n)| k_spring = (rho_ua / rho_scm) * THZ * PHI_RES beta(t) from derive_beta_i or 0.603 * |E| * Z * cos(pi*t) (integrator) ```**

UbiForceBalanceIntegrator (MAIN_1_CoAnQi.cpp:2852) ```cpp static double computeUbi(...) { return 0.603 * abs(E_single) * Z * cos(M_PI * t_norm); } static double applyForceBalance(Ug_sum, Um, params) { return Ug_sum - Ubi + Um + dissipation; } ``` Applied to 22 Tier1/Tier2 apps (UnifiedField_Ug1-Um, CompressedMUGE, ResonanceMUGE).

Live Convergence (2026-05-27, DERIVATIONS only) ``` r_hz (M=1e30, t_n=0, rho=633333.333, beta=0.65, G=0.02948) = 1.7095376216580647e+19 m QuantumChain rho_energy direct = 633333.3333333334 DERIVATIONS condensed = 633333.333 (exact) FUBi + FUBii root residual = 0.0 F_U residual < 1e-10 (all solvers) ```

Cross-Solver r_hz / F_U=0 Identity Table | Layer / Engine | r_hz (m) | FUBi+FUBii | F_U | rho / beta / G source | Ref | |--------------------------------

--------|--------------------|------------|----------|--------------------------|-----| | DERIVATIONS.derive_habitable_zone_radius | 1.7095376216580647e+19 | 0.0 | <1e-10 | 10 derive_* + QuantumChain | _uqff_primitives:766 | | QCalcGeom v2.1.0 HabitableZoneCalculator | identical order | 0.0 (1e-12) | <1e-10 | DERIVATIONS wired CP4 | QCalcGeom:458,524 | | CP4 Ubi corrections (FUBi/FUBii+Um) | identical | 0.0 | <1e-10 | CP4 = Ubi layer | uqff/CP4.py + QCalcGeom | | UbiForceBalanceIntegrator 22 apps | per-app crossing | 0.0 | 0.0 | 0.603 * |E|*Z*cos (MAIN) | MAIN_1:2878 | | Simultaneous7LayerSolverBridge (v1.5) | consistent | 0.0 | <1e-10 | 7 layers + explicit Ubi | UQFFAtomicSolverIntegration.py | | 60/60 QCalcGeom tests (T81-T87 UBS) | all converge | 0.0 | <1e-10 | only derived constants | QCalcGeom: run_qcalcgeom_tests |

CP1->CP4 Pipeline (QCalcGeom + uqff/CP*.py) ``` CP1: raw vacuum (RHO_VAC_SCM, ratio=10 from dpm v3.0) CP2: scaled (mass 12.5, density laws, 10.0 ratio) CP3: resonance/quantum (CompressedMUGE/ResonanceMUGE) CP4: Ubi corrections (FUBi/FUBii + Um via integrator) -> F_U + solve_habitable_zone All layers now consume DERIVATIONS (no CODATA/hardcoded post #8-9 fidelity) ```

Cross-References - QCalcGeom.py:14 (simultaneous Eq1/Eq2) - MAIN_1_CoAnQi.cpp:2852 (integrator + 22 apps) - _uqff_primitives.py:766 (derive_hz root solve) - dpm_vacuum_manifold.py:6 (F_U from Ug_family + FUBi/FUBii) - UQFFAtomicSolverIntegration.py (7-layer bridge) - LEDGER_FIRST_PRINCIPLES_DERIVATIONS.md:85 (F_U = Ug_sum - Ubi + Um) - WhitepaperGapAnalysis.md:102 (G5/G6 gap closure via simultaneous solvers)